

ShopSphere Customer Intelligence System

Final Report — Retention, Value and Experimentation

Data window: 2024-06 – 2026-06

Generated: 2026-07-11 · Headline KPIs queried live from the MySQL gold layer; model metrics quote the validated notebooks

All data synthetic, calibrated to 2026 industry benchmarks

Executive Summary

ShopSphere is a synthetic-but-realistic online retailer analyzed end-to-end: 24 months, 12,000 customers, ~2.36M rows through a bronze → silver → gold MySQL warehouse. The central finding: **69.2% of buyers never purchase a second time**, and the highest-leverage actions all attack that number.

KPI	Value	KPI	Value
Total completed revenue	\$1,664,813.13	One-time buyer rate	69.2%
Completed orders	17,811	Cart abandonment	70.6%
Average order value	\$93.96	Top-20% buyers' revenue share	56.9%
Unique buyers	11,323	Blended paid CAC	\$73.89
NPS (response-weighted)	45.5	A/B free-shipping lift	+15.1% (p = 7.9e-04)

Top recommendations (detail in §6): roll out the free-shipping threshold (+15.1% conversion, p = 7.9e-04); run a second-purchase program for the 70% one-time buyers; win back the Hibernating segment (30% of buyers, 18% of revenue); protect Champions (4% of buyers, 12% of revenue).

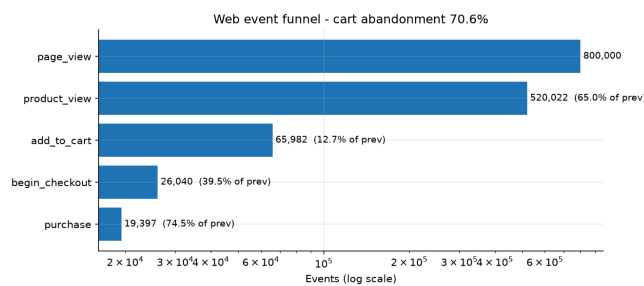
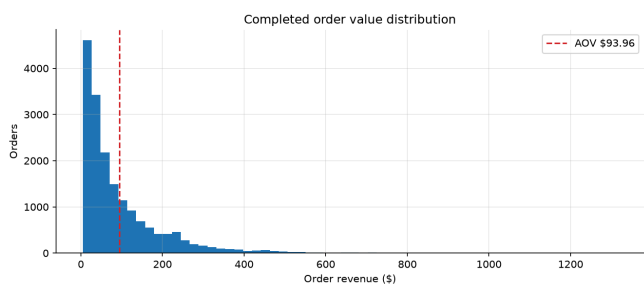
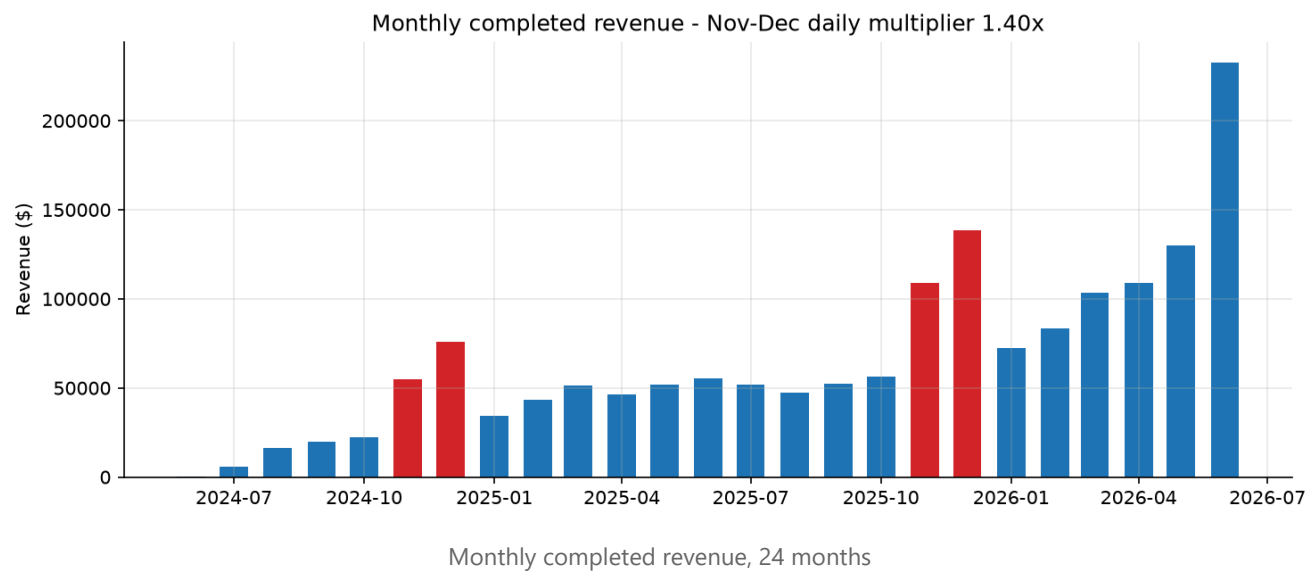
1 · Data Foundation — Trust Before Analysis

Every number in this report sits on a verified pipeline:

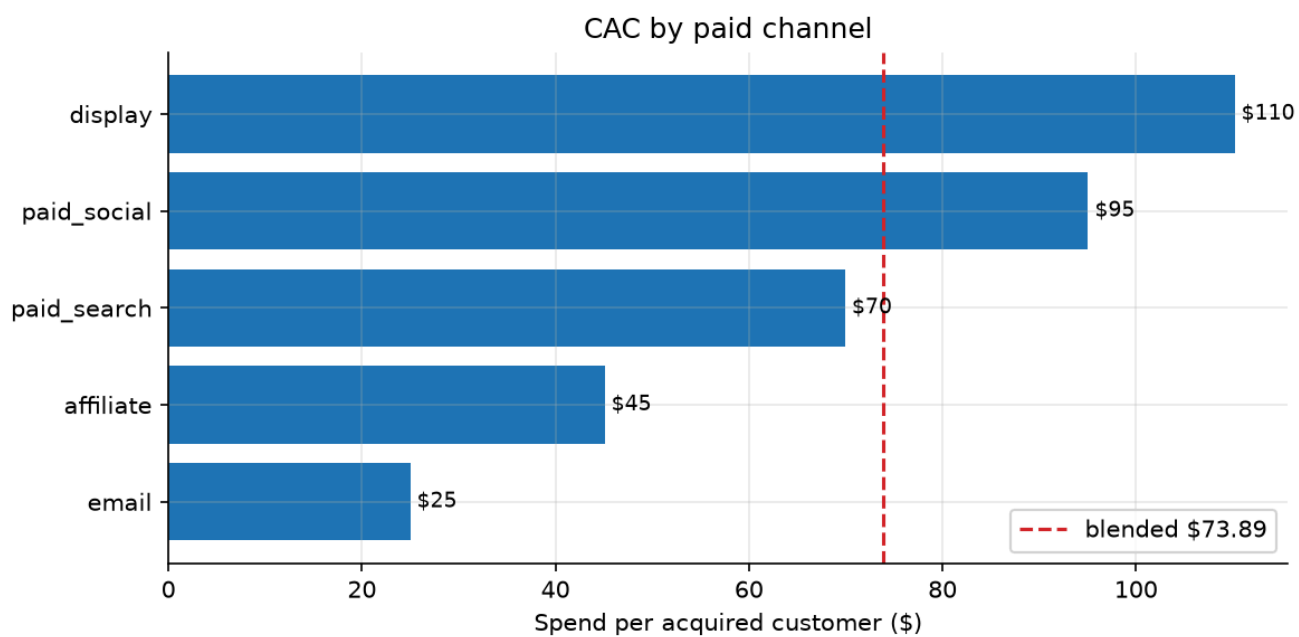
- **Seeded generation** — 8 behavioral realism rules (Pareto concentration, repeat-purchase ladder, funnel decay, channel economics, seasonality, price-elastic baskets, satisfaction–loyalty link, a real A/B effect). A calibration gate fails the build if any of 8 checks drifts from its industry band: **8/8 PASS**.
- **Deliberate dirt, graded cleaning** — 9 defect classes injected with a ground-truth manifest; the bronze → silver cleaning pipeline was graded against it: **9/9 caught (PERFECT)**. 2,362,710 bronze rows → 2,362,203 silver rows; e.g. 290 duplicate orders removed (19,687 → 19,397).
- **SQL gold layer** — 12 views (KPIs, window functions, cohort retention). Gate vs independent pandas recomputation: **5/5 PASS**, cohort matrix exact in 267/267 cells.

2 · Business Health

Revenue grows steadily with a 1.40× Nov–Dec seasonal peak; the final month's jump is the growth curve's in-month ramp, not an anomaly. AOV averages \$93.96 across 17,811 completed orders.



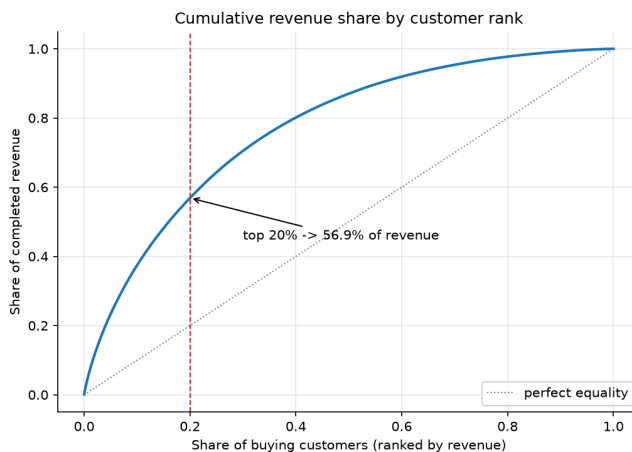
The funnel loses 70.6% of carts before checkout — the single largest recoverable revenue pool. On the acquisition side, blended paid CAC is \$73.89 with email converting ~3× paid social.



Customer acquisition cost by channel

3 · Who the Customers Are

Revenue is concentrated: the top 20% of buyers contribute 56.9% of revenue, and 69.2% of buyers purchased exactly once. The repeat-purchase ladder confirms the classic pattern — the hardest step is first → second order; each later step retains more.



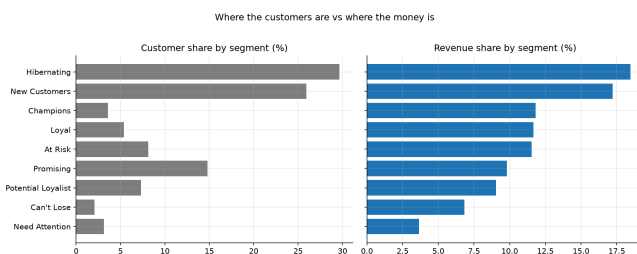
Cumulative revenue vs buyer rank



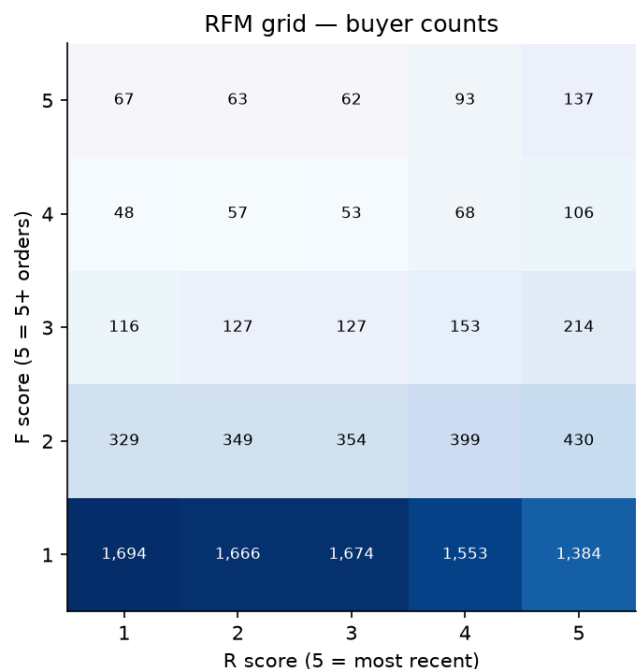
Repeat-purchase ladder

RFM segmentation (11,323 buyers, 9 segments)

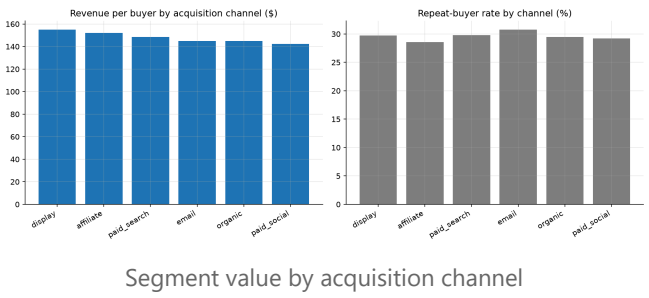
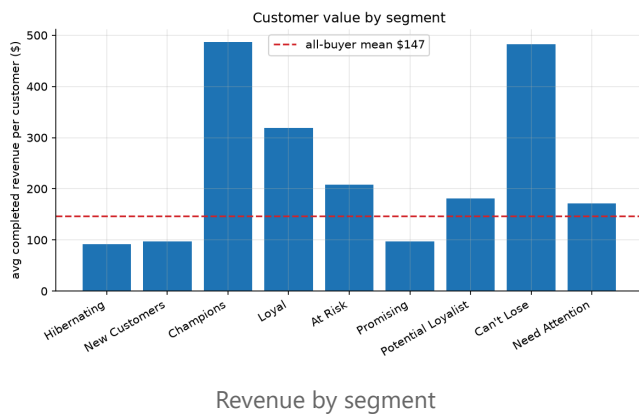
Champions are 4% of buyers but 12% of revenue (avg \$487 vs \$147 overall); Hibernating is the largest segment at 30% of buyers holding 18% of historic revenue — the natural win-back target.



Segment sizes

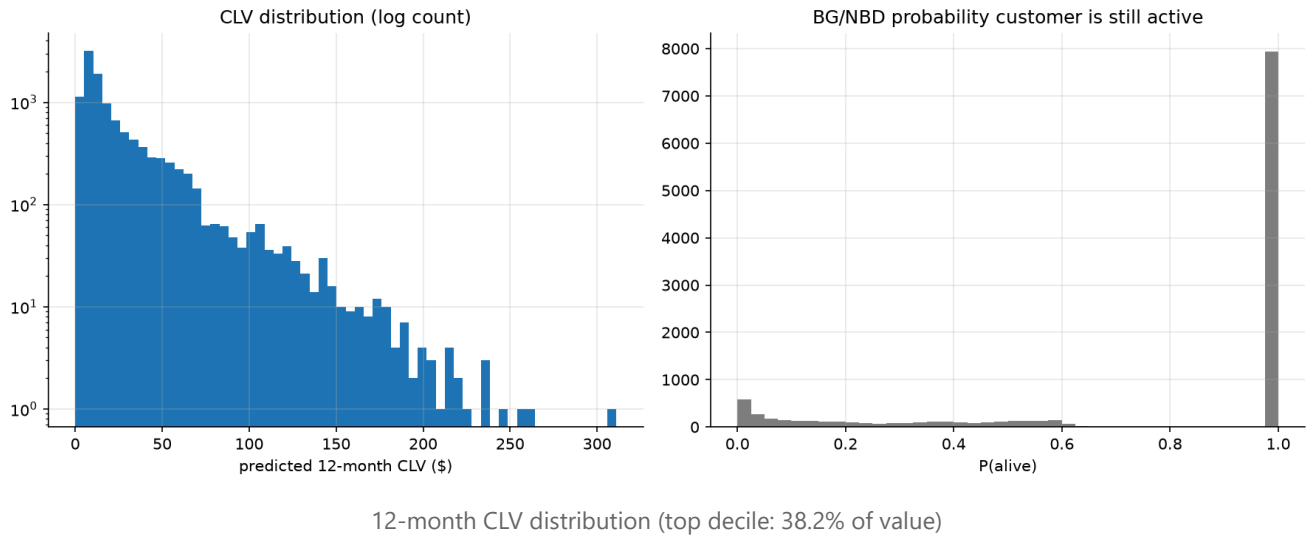


R × F grid

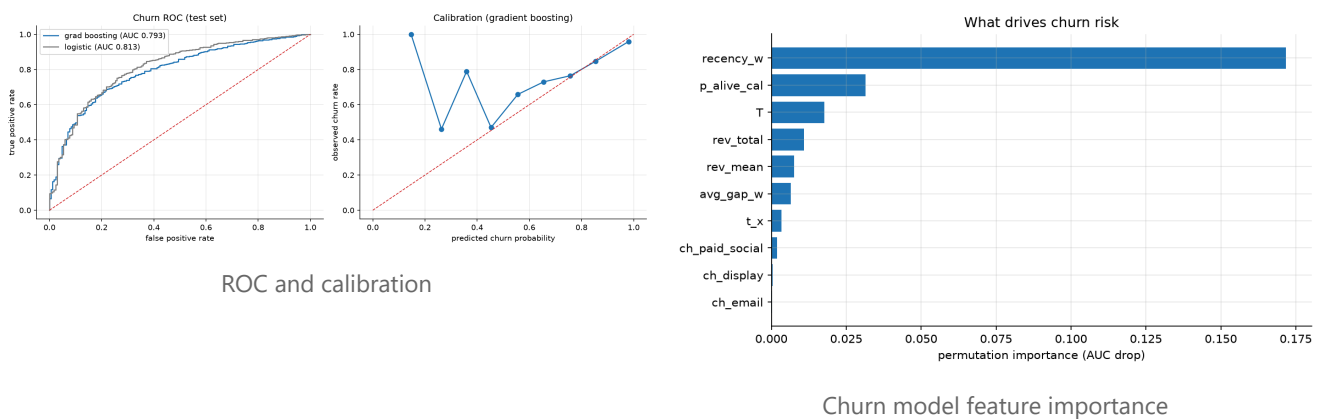


4 · Value & Risk — CLV and Churn

A BG/NBD + Gamma-Gamma model (implemented on scipy, validated by a 26-week time-split backtest: predicted/actual repeat purchases = 1.06) projects **\$282,875 of 12-month CLV** across the base — mean \$25.09 per buyer, with the top decile holding 38.2% of it. Future 12-month CLV alone does not repay the \$73.89 blended CAC; acquisition must be justified on multi-year value and aimed at high-CLV lookalikes.

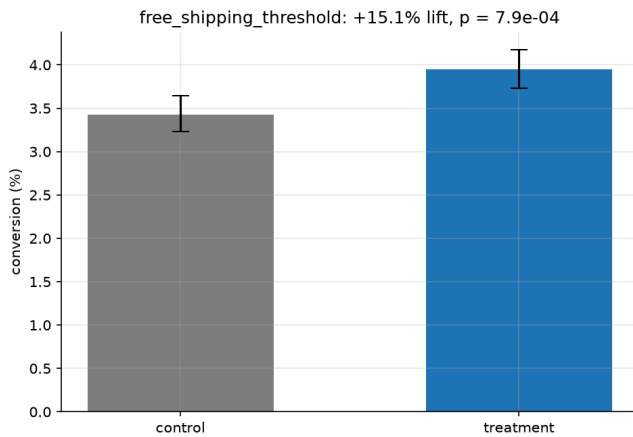


A churn model (26-week holdout, leakage-checked features) reaches AUC 0.813 against an 89.9% base churn rate — good enough to rank buyers for retention spend, with recency and frequency dominating the signal.

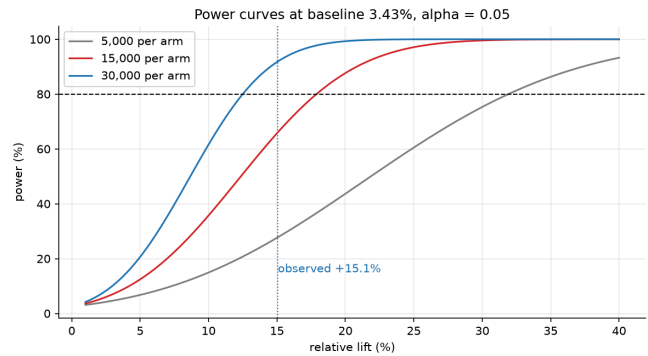


5 · Experimentation — Free-Shipping Threshold

A 30,000-per-arm A/B test (randomization verified: SRM χ^2 $p = 1.0000$) raised conversion from 3.43% to 3.95% — **+15.1% relative lift** (95% CI +6.0% ... +24.9%, $z = 3.36$, $p = 7.9\text{e-}04$), at 91.9% achieved power against a +12.5% design MDE.



Conversion by arm with 95% CI



Power curve vs relative lift

Guardrail: converted-order revenue was \$88.80 (treatment) vs \$97.36 (control), Welch $p = 0.071$ — a suggestive but not significant basket shrink. Revenue per assignment still favors treatment (\$3.17 vs \$3.00). Ship, and monitor basket size in production.

6 · Recommendations

1. **Roll out the free-shipping threshold to 100% of traffic.** +15.1% conversion at $p = 7.9e-04$ with power 91.9%; keep the basket-size guardrail on a dashboard (it was suggestive at $p = 0.071$, not proven).
2. **Stand up a second-purchase program.** 69.2% of buyers stop at one order, and the ladder shows first → second is the hardest step; a post-first-purchase sequence attacks the villain number of this whole analysis.
3. **Win back Hibernating, protect Champions.** Hibernating = 30% of buyers / 18% of revenue (reactivation offers); Champions = 4% / 12% at \$487 avg (loyalty perks, no discounting needed).
4. **Target acquisition by CLV, not volume.** Mean 12-month CLV (\$25.09) is far below blended paid CAC (\$73.89); the top CLV decile holds 38.2% of future value — shift budget toward email/lookalikes of high-CLV profiles and away from the most expensive paid channels.
5. **Attack cart abandonment (70.6%).** The funnel's largest leak; recovery emails and checkout friction reduction are the standard levers, now measurable month-over-month in the gold layer.

Appendix · Method & Reproducibility

- **Stack:** Python (seeded generators, cleaning, scipy/sklearn models) → MySQL 8 medallion warehouse (bronze/silver/gold) → SQL window functions → Excel workbook, Power BI dashboard, this report.
- **Determinism:** one seed in `settings.py`; same seed ⇒ byte-identical CSVs.
- **Gates** (all green at build time): calibration 8/8, cleaning quality 9/9, gold KPI gate 5/5, Excel workbook gate 5/5, Power BI DAX acceptance 8/8, pytest suite, and this report's own self-validation gate (KPIs re-checked against live SQL after render).
- **Headline KPIs in this report are queried live** from the gold views at build time with the same SQL as the M3 gate; model metrics quote the executed, validated notebooks.

ShopSphere is a synthetic retailer: all data generated, calibrated to 2026 industry benchmarks, and verified by the gates above.
Report built by `python/05_delivery/build_final_report.py`.